

Structured Divergence in AI Creative Generation: A Same-Input Study of Puzzle Hunt Pipeline Outputs

[Author]
[Institution]
USA
[email]

Abstract

AI creative generators are stochastic: the same pipeline, given the same inputs, produces different outputs across independent runs. Practitioners deploying these systems lack principled guidance on which design decisions will be stable across runs—and can therefore be trusted on a single execution—and which will vary, requiring multiple runs and human selection to control quality.

We address this problem through a *same-input divergence* study: two fully independent, clean-room runs of a complete AI puzzle hunt generation pipeline from identical inputs—the same Age of Empires II world data, scope document, and toolkit specification—compared at five levels of abstraction (structural, pool, answer-word assignment, extraction mechanism, and micro-design). An AI expert panel evaluated both outputs on a 7-dimension, 45-point weighted rubric using identical reviewer configurations.

The results reveal a structured, not random, pattern of divergence. Every decision explicitly specified in the creative brief—scale (5 feeders + 1 meta), narrator voice, audience, format, and solve-time target—is identical across both runs. Every decision left to the pipeline diverges substantially: round count (1 vs. 3), meta mechanism family (crossword grid vs. vowel-count index followed by anagram resolution), and answer words (Jaccard = 0.00 across all six words including the meta). We identify a distinct phenomenon, *domain semantic gravity*: despite zero lexical overlap, 3 of 5 feeder slots yield domain-adjacent word pairs (e.g., ONAGER/SIEGE; PATROL/MARCH), revealing that the domain exerts a semantic basin of attraction without constraining specific word choice. Panel quality scores are stable at the pass/fail level (all authored puzzles pass in both runs; normalized aggregate scores within 3 percentage points) while showing a systematic +1.2-point uplift in Elegance and Reading Reward in the second run, consistent with a genuine mechanism-level improvement.

We introduce same-input divergence as a generalizable experimental paradigm for characterizing AI pipeline reliability, establish *specification drives convergence* as the primary practical principle—every under-specified decision is a variable decision—and derive actionable deployment guidance for practitioners operating AI creative pipelines at scale.

ACM Reference Format:

[Author]. 2026. Structured Divergence in AI Creative Generation: A Same-Input Study of Puzzle Hunt Pipeline Outputs. In . ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Two independent runs of the same AI puzzle hunt pipeline, fed identical inputs — the same Age of Empires II world data, the same scope document, the same toolkit specification — produced hunts that share zero answer words (Jaccard similarity: 0.00), use categorically different meta mechanisms (a crossword grid trace versus a vowel-count index followed by anagram resolution), adopt different round architectures (a single flat round versus a three-round gated structure), and arrive at different meta answers (WOLOLO versus GRAIL). At the same time, both hunts maintain identical narrator voice, identical audience specification, identical format, identical total puzzle count, and similar solve-time estimates. Both pass the same quality threshold under AI expert panel review with normalized scores within four percentage points of each other. The divergence is real, total, and systematic at some levels of the pipeline; the convergence is equally real, complete, and systematic at others.

This pattern poses a practical problem for practitioners deploying AI creative generators. Current pipelines for AI-assisted creative production — whether for puzzle hunts, tabletop campaign design, interactive fiction, or procedural game content — yield outputs that vary stochastically across runs. Practitioners face a binary choice in response: review every output decision manually (expensive, and defeats the purpose of automation) or trust the pipeline’s outputs as-is (cheap, but potentially arbitrary where it should not be). Neither strategy is principled, because neither is informed by a characterization of which decisions are actually stable across runs and which are genuinely variable. Without this characterization, practitioners cannot know which outputs to run once and trust versus which to run multiple times and select from. They either over-invest human oversight where the AI is reliable, or under-invest where reliability is absent. A principled account of divergence structure is the necessary foundation for rational deployment of AI creative pipelines.

We address this problem through a *same-input divergence* experiment: two fully independent runs of a complete puzzle hunt generation pipeline from identical input documents, compared at five levels of abstraction — structural decisions, puzzle pool selection, answer word assignment, extraction mechanism design, and micro-level clue formulation. The experiment treats stochasticity not as a confound to be eliminated (as in reproducibility research) but as an instrument for revealing the structure of the pipeline’s design space. Decisions where the two runs converge,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference'17, Washington, DC, USA

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

despite stochastic generation, are decisions where the design space is constrained enough that independent generative agents reach similar conclusions. Decisions where the two runs diverge are decisions where the design space offers many equally valid options and nothing in the specification privileges one over another. The pattern of convergence and divergence, read across five levels, characterizes which parts of the pipeline are reliable and which are not.

Our central empirical finding is that divergence in this pipeline is structured, not random, and that the structure is predicted by a single principled distinction: *specification drives convergence*. Every decision that was explicitly specified in the creative brief — scale (five feeders plus one meta), narrator voice (the Monk persona), output format (printable Markdown), target audience (solo AoE2 player, casual), and solve-time target — is stable across runs to the point of identity. Every decision left to the pipeline — round count, round themes, meta mechanism family, answer word selection, extraction logic within puzzle types, clue formulations — diverges substantially or totally. Notably, this is not a domain-knowledge effect: the Age of Empires II domain is rich and well-defined, but domain richness neither constrains the round structure nor determines the meta answer. Domain content acts instead as what we call *semantic gravity* at the word-selection level: even when answer word sets are completely disjoint as strings, three of five feeder slots contain domain-adjacent pairs (LOOM / FORGE from the technology domain; ONAGER / SIEGE from siege warfare; PATROL / MARCH from unit commands), suggesting that the content library exerts a partial anchoring force on word choice even when it does not constrain which word is chosen. Domain knowledge is neither necessary nor sufficient for stability; specification is both.

This paper makes five contributions. First, we introduce the same-input divergence paradigm as a general experimental method for characterizing the stability structure of any AI creative pipeline — a complement to prompt sensitivity analysis (which varies the input to measure output sensitivity) that instead fixes the input and characterizes natural output variation across runs. Second, we provide concrete empirical data on stable and variable decisions at five levels of abstraction in a complete AI puzzle hunt pipeline, grounding the stability analysis in the specific numerical results of the AoE2 comparison: structural and assignment-level divergence is near-total on exact-string metrics; quality-level convergence is near-complete on pass/fail and normalized aggregate scores. Third, we establish the specification-drives-convergence finding as the primary explanatory principle: stability tracks what the brief said, not what the domain constrains, in this pipeline. Fourth, we characterize domain semantic gravity as a distinct phenomenon that operates below the level of specification but above the level of random variation, anchoring word-choice distributions without fixing individual words. Fifth, we derive actionable practical recommendations from the stability data: run once and trust for explicitly specified decisions; run multiple times and select for unspecified ones; use specification completeness as the primary lever for controlling pipeline reliability.

2 Related Work

Our study sits at the intersection of four research areas: quality-diversity and diversity in procedural content generation, stochasticity and prompt sensitivity in language models, reproducibility in generative systems, and design space theory. We situate same-input divergence within each area and clarify the experimental and conceptual distinctions that make our contribution novel.

2.1 Quality-Diversity and Diversity in Procedural Generation

A significant strand of computational creativity and procedural content generation (PCG) research treats diversity as a primary design objective. The MAP-Elites algorithm of [?] exemplifies this orientation: it maintains an archive of solutions partitioned by behavioral descriptors, explicitly maximizing both quality within each niche and variety across niches. This quality-diversity (QD) framing has been widely adopted in game content generation [?], where diverse outputs are desirable to avoid player boredom and cover the space of possible experiences.

Diversity measurement in PCG follows a complementary research agenda. [?] propose expressive range analysis, a framework for visualizing and quantifying the variety of outputs a generator can produce across dimensions such as linearity and leniency. [?] apply evolutionary search to generator parameters with diversity as a fitness component, treating expressiveness as a property to be maximized through search.

Our work differs from this tradition in a fundamental respect. The QD and expressive range literatures ask: *how much diversity can this generator produce, and how do we maximize or measure it?* We ask a different question: *given a fixed input specification, which components of the output are diverse and which are not, and why?* We do not seek to maximize divergence; we seek to characterize its structure. The distinction is between diversity as a goal to engineer and divergence as a phenomenon to explain. Understanding this structure has immediate consequences for how practitioners deploy AI creative generators in production pipelines, because it determines where human oversight is necessary and where a single AI run can be trusted.

2.2 Stochasticity and Prompt Sensitivity in Language Models

Language models introduce stochasticity through temperature-controlled sampling, and several lines of research examine how this stochasticity shapes output variation. [?] study the effect of stylistic parameters, including temperature, on the diversity and quality of generated text, establishing that temperature is a primary lever for controlling output variation. This work treats temperature as a design parameter to be set; our study treats temperature as fixed and asks what remains variable despite that setting.

A related and more recent body of work examines prompt sensitivity: the observation that LLM outputs vary substantially with small changes to the input prompt. [?] demonstrate that different phrasings of the same factual query yield dramatically different model responses, coining the term “prompting” as a search problem. [?] show that LLM classification is sensitive to label ordering and demonstration selection in few-shot settings, and [?] extend this

to show that permutation of in-context examples strongly affects output quality, with optimal orderings differing across models and tasks.

These prompt sensitivity studies vary the input to measure how sensitive the output is to input perturbation. Our experimental paradigm is precisely the inverse: we *fix the prompt* and measure output variation across independent runs. This targets a distinct phenomenon — the natural stochasticity inherent in a generative model at fixed temperature, rather than sensitivity to prompt design. Moreover, our analysis operates at the level of a structured, multi-stage creative artifact (a puzzle hunt with rounds, answer words, mechanisms, and clue formulations), not at the level of a single model response. This multi-level analysis is necessary to ask which *components* of the output are sensitive to stochasticity, a question that single-response studies cannot address.

2.3 Reproducibility in Generative Systems

A third relevant area is reproducibility in machine learning and generative systems. ?] provide a comprehensive framework for reproducibility in ML research, emphasizing seed control, hyperparameter reporting, and environment specification as prerequisites for replicable results. The general thrust of this literature is that stochasticity is a confound to be controlled: reproducibility is achieved when two runs of the same system produce the same output.

In computational creativity, ?] examine reproducibility as one criterion for evaluating creative systems, raising the question of whether a system's outputs are predictably related to its inputs. In PCG specifically, seed control is standard practice: fixing the random seed guarantees that a procedural generator produces identical output on repeated runs, which is essential for debugging, level handcrafting, and player-facing consistency.

Our study takes the opposite stance toward stochasticity. Rather than eliminating or controlling it, we deliberately allow it to operate and use the resulting variation as data. Our goal is not to make runs identical but to understand *which decisions are inherently stable across runs and which are inherently variable*. Stable decisions are those where the design space is sufficiently constrained that independent runs converge to the same choice; variable decisions are those where the design space offers many equally valid options and independent runs do not converge. This framing converts stochasticity from a nuisance to be suppressed into an instrument for characterizing the structure of the design space itself.

2.4 Design Space Theory and Constrained Creative Decisions

The stable/variable distinction we observe empirically connects to a principled theoretical tradition in design cognition. ?] characterizes design as search through a space of possibilities under constraints: the designer's task is to find an acceptable point in a large space of potential solutions, guided by problem-structure and satisfaction criteria. The size and structure of the relevant design space determines how constrained any particular decision is.

?] extends this framework with empirical observations of design cognition, distinguishing between problem-focused strategies (working from constraints inward to solutions) and solution-focused strategies (proposing solutions and testing them against constraints). Critically, this work shows that experienced designers quickly identify which aspects of a design problem are heavily constrained (leaving little creative latitude) and which are relatively free (requiring deliberate creative choice). The stable decisions we observe — round structure, puzzle type variety, meta mechanism type — are analogous to heavily constrained decisions in Lawson's framework: domain knowledge, audience requirements, and structural specifications together reduce the space of valid options to a small set, and independent agents navigating that space converge on similar choices. The variable decisions — specific answer words, extraction mechanism details within a puzzle type, individual clue formulations — correspond to design-space-free decisions: many options are equally valid, no constraint privileges one over another, and independent agents diverge.

?] formalizes this distinction in the function-behavior-structure model of design, which separates the functional requirements a design must satisfy (analogous to our domain constraints), the behavioral properties it exhibits (observable outputs), and the structural decisions that produce those behaviors (analogous to our design-space choices). Structural decisions that are over-determined by functional requirements leave little latitude and tend to be stable; structural decisions with many behavioral-equivalent options are inherently variable. Our empirical findings in a large-scale AI creative pipeline provide concrete, quantitative support for this theoretical prediction.

2.5 Same-Input Divergence as a Computational Creativity Criterion

Characterising the reliability structure of a creative pipeline — identifying which generative decisions converge across independent runs and which diverge — is a prerequisite for evaluating that pipeline as a creative system under established CC frameworks. The SPECS framework of ?] provides a standardised procedure for CC evaluation, requiring that evaluation criteria be applied consistently and that a system's outputs be assessable as creative artefacts. This evaluation is only possible if we can distinguish systematic output properties from stochastic noise: a system whose every run produces a categorically different meta mechanism (Level 1) and zero shared answer words (Level 3) cannot be assessed under criterion C8 ("the system reliably produces creative artefacts") without first characterising which decisions drive that instability. By contrast, a system where panel scores are stable across runs (Level 6) and pass/fail verdicts converge satisfies C8 in the relevant dimension. Same-input divergence operationalizes this criterion: it makes C8 empirically testable by decomposing pipeline reliability into per-decision measurements.

This framing connects to ?]'s evaluative criteria for creative systems, which require that artefact quality be reproducible across independent runs for quality assessments to be meaningful. Ritchie's criteria cannot be applied if the relationship between input specification and output quality distribution is unknown. The same-input divergence paradigm provides the measurement instrument needed

to establish this relationship: by running the pipeline twice from identical inputs and comparing outputs at multiple abstraction levels, we identify which quality dimensions are stable (and therefore assessable as reliable system properties) and which are contingent on incidental generative choices. We make no claim that divergence is evidence of creativity, nor that convergence implies its absence. Rather, we argue that characterising the structure of same-input divergence is a prerequisite for any CC evaluation of a multi-stage generative pipeline — you cannot evaluate creative reliability without first measuring it.

Gap Statement

Taken together, these four research areas provide partial predecessors but no direct treatment of the problem we address. Quality-diversity work maximizes or measures diversity as a goal but does not ask where divergence originates in a multi-stage pipeline or why some stages converge while others do not. Prompt sensitivity work varies inputs to measure output sensitivity but does not fix the input and characterize natural stochasticity across a structured creative artifact. Reproducibility work suppresses stochasticity rather than characterizing its structure. Design space theory provides the conceptual vocabulary for constrained vs. unconstrained decisions but has not been connected empirically to the behavior of LLM-based creative generators.

No prior work characterizes the structure of same-input divergence at multiple levels of abstraction in an AI creative pipeline. Existing work either maximizes diversity, controls it, or measures it holistically without decomposing where in a multi-stage generative process it originates and which stages are responsible. Our study fills this gap with an empirical same-input experiment, a five-level comparison methodology, and a principled explanation of the resulting divergence structure grounded in design space theory.

3 Methodology

3.1 The Same-Input Divergence Paradigm

Same-input divergence refers to the natural variation that arises when an AI creative pipeline is executed twice from identical inputs. In our experimental instantiation, the inputs held constant across both runs are: the domain knowledge base (world data files describing Age of Empires II civilizations, units, technologies, maps, and economy), the scope document specifying hunt scale, audience, and format, the pipeline specification (the puzzle hunt toolkit skill library and its design principles), and the reviewer profile library used for AI panel evaluation. The only source of variation between the two runs is LLM stochasticity at fixed temperature: the same sequence of generative prompts, applied to the same context, producing outputs that differ because the underlying language model samples from its conditional distribution independently on each run.

Three experimental paradigms are superficially similar to same-input divergence and must be distinguished from it. *Reproducibility studies* [?] seek to eliminate divergence by fixing random seeds or controlling all sources of stochasticity; their goal is identical outputs across runs, and divergence is treated as an experimental failure. Our goal is the inverse: we deliberately allow stochasticity to operate and use the resulting divergence as data. *Ablation studies*

vary one component of the input pipeline at a time to measure its causal effect on outputs; our paradigm varies nothing in the input and instead characterizes the variation that arises endogenously from the generative process itself. *Sensitivity analysis* [?] varies inputs (typically prompt phrasing) to measure output sensitivity to those variations; again, our paradigm inverts this by fixing all inputs and asking which output decisions are robust to stochastic variation and which are sensitive to it.

The central question of same-input divergence is not *how much* divergence a generator produces across runs — a question that quality-diversity metrics answer — but *where* in the generative process divergence originates and *why* some decisions converge across independent runs while others do not. This question is meaningful only when analyzed at multiple levels of abstraction within the generated artifact, which motivates the five-level comparison framework described in Section 3.3.

3.2 Pipeline and Corpus

Our corpus consists of two independently generated puzzle hunts produced by the same AoE2 knowledge base, scope document, and generation toolkit.

Hunt 1 (“Wololo”). Hunt 1 is the primary Age of Empires II puzzle hunt scenario generated and fully documented in the companion generative pipeline paper [?]. It comprises five feeder puzzles and one meta-puzzle, with meta answer WOLOLO — the distinctive sound of the in-game Monk unit performing a conversion. The meta mechanism is a crossword-style extraction: five feeder answers (CASTLE, ONAGER, LOOM, TOWER, PATROL) each contribute one letter via a crossword grid, spelling the meta answer. All pipeline stages were completed and all five puzzles were authored and tested. AI panel scores for all five puzzles are available from the panel review stage (Stage 8), using the C8 panel of three expert reviewer profiles and a 7-dimension rubric scored on a 45-point scale.

Hunt 2 (“Run 2”). Hunt 2 was generated under clean-room conditions from the same inputs as Hunt 1. The world data files (civs.md, units.md, techs.md, maps.md, economy.md), SCOPE.md brief (a 5-puzzle AoE2 hunt for a casual solo audience, estimated 2–3 hours solve time), toolkit skill library, and 29-profile reviewer panel library are identical to Hunt 1. The generating agent was explicitly instructed not to access Hunt 1’s PUZZLE-POOL.md, ROUNDS.md, PUZZLES.md, or puzzles/ directory at any stage; generation of the candidate pool, round structure, meta design, and individual puzzles proceeded without any Hunt 1 artifacts in context. This clean-room isolation ensures that any convergence observed between the two hunts reflects genuine structural constraint rather than imitation, and any divergence reflects genuine stochastic variation rather than deliberate differentiation.

Hunt 2 produced five feeder puzzles and one meta-puzzle, with meta answer GRAIL. The feeder answers are FORGE, MARCH, SIEGE, FAITH, and RELIC. The meta mechanism is categorically distinct from Hunt 1: for each feeder answer, the solver counts the vowels in the answer word and adds one, obtaining an index; the letter at that index position is extracted; the five extracted letters (R, A, G, I, L) are anagrammed to yield GRAIL. Three of the five feeder puzzles (P1–P3) were fully authored and submitted for AI panel review; P4

and P5 are structurally specified but not yet authored at the time of analysis. Panel scores for P1–P3 are available and are included in the panel score comparison analysis.

Both hunts used the same 29-profile v2 reviewer library for AI panel evaluation, the same 7-dimension rubric, and the same three-reviewer C8 panel configuration (profiles corresponding to the Katz, Snyder, and Young reviewer personas).

3.3 Five-Level Comparison Framework

We compare the two hunts at five levels of abstraction, proceeding from the most coarse-grained structural properties to the most fine-grained micro-design decisions. This multi-level architecture is necessary because divergence may be present at some levels and absent at others; a single aggregate similarity score would obscure precisely the structure we seek to characterize. Table 1 summarizes all five levels, their comparison units, metrics, and stability thresholds.

Level 1 — Structural. The unit of comparison is the hunt as a whole. Metrics are: round count (exact match), meta mechanism category (categorical classification: crossword extraction, vowel-count index extraction, acrostic, anagram, etc.), and total puzzle count. This level asks whether the two hunts converge on the same overall architecture.

Level 2 — Pool. The unit of comparison is the 15-puzzle candidate pool generated in Stage 3 of the pipeline. Metrics are: puzzle type distribution (count per canonical type, with KL divergence as a summary statistic), overlap between the five puzzles selected from each pool (Jaccard similarity on 5-element sets), and score distribution comparison across the pool (mean, variance, and range of AI panel scores for the full candidate set).

Level 3 — Assignment. The unit of comparison is the answer word set. Metrics are: exact word overlap (Jaccard similarity on the 5-word feeder sets), semantic similarity of the two answer word sets computed as the mean pairwise cosine similarity of word embeddings after optimal bipartite matching, and meta feeder overlap (the subset of feeder answers that appear in both hunts).

Level 4 — Mechanism. The unit of comparison is the puzzle-level extraction method within each puzzle type. Each puzzle is manually coded for mechanism type (e.g., chain-following, property identification, grid traversal, sequence ordering) and mechanism variant within type. Metrics are: categorical mechanism type per puzzle slot, mechanism type distribution comparison across the five-puzzle set, and a mechanism similarity matrix computed from the coding.

Level 5 — Micro-design. The unit of comparison is individual clue formulations, flavor text, and grid content. After alignment of comparable puzzle slots by puzzle type, metrics are: character-level edit distance on equivalent clue sections, and semantic similarity of clue sets computed as cosine similarity on sentence embeddings of the full clue text per puzzle.

3.4 Stability Classification

A design decision is classified as *stable* if both independent runs produce the same or functionally equivalent output for that decision. A decision is classified as *variable* if the outputs differ in ways

Table 1: Five-level comparison methodology. Each level specifies a unit of comparison, the metrics applied, the stability threshold, and the predicted direction (stable or variable) based on design-space theory.

Level	Unit	Metric(s)
1 – Structural	Hunt (whole)	Round count (exact); meta type (categorical)
2 – Pool	Candidate pool	Type KL divergence; Jaccard (top-5); score dist.
3 – Assignment	Answer word set	Jaccard (5-word sets); embedding cosine sim.
4 – Mechanism	Puzzle mechanism	Mechanism type coding; distribution comparison
5 – Micro	Clue formulations	Edit distance; sentence embedding cosine sim.

that represent a genuine design choice rather than a surface paraphrase. Stability is assessed per level using the thresholds specified in Table 1; these thresholds are calibrated to domain properties rather than set at arbitrary conventional values.

At Levels 1 and 2 (structural and pool), stability is assessed against exact-match and distributional thresholds: round count must exactly match; meta mechanism type must be from the same categorical class; puzzle type KL divergence must fall below 0.5 nats (corresponding to a distribution whose most probable type receives no more than twice the probability mass it would under the other run’s distribution); and pool Jaccard similarity must exceed 0.4 (corresponding to more than one-third overlap in the top-5 selections from a 15-candidate pool).

At Level 3 (assignment), a Jaccard threshold of 0.4 on the 5-word answer set is again used, and semantic similarity via embedding cosine distance provides a soft similarity measure that accounts for thematically related but lexically distinct answer words (e.g., CASTLE and SIEGE are distinct words but thematically proximate in the AoE2 domain). A decision at Level 3 is classified as variable if answer word Jaccard falls below 0.4 and the embedding similarity score places the two answer sets in different regions of the semantic space.

At Levels 4 and 5 (mechanism and micro-design), stability requires that the dominant mechanism type assigned to comparable puzzle slots matches across the two runs. A variable classification is issued when different mechanism types are used for structurally analogous puzzle slots (e.g., when one run uses chain-following extraction for its civilizations puzzle and the other uses property-grid identification). At Level 5, semantic similarity of clue formulations is expected to be low in all cases, and the analysis focuses on characterizing the distribution and range of micro-design variation rather than a binary stable/variable classification.

3.5 Panel Score Comparison

Panel scores for Hunt 2’s three authored puzzles (P1, P2, P3) were generated using the same C8 panel (Katz, Snyder, Young reviewer profiles) and the same 7-dimension, 45-point rubric applied to Hunt 1 in the companion AI panel paper [?]. Each puzzle is scored across seven dimensions: clue clarity, fairness of extraction, domain knowledge integration, difficulty calibration, thematic coherence, aha-moment quality, and solver experience. Dimension scores sum to a total panel score out of 45.

Score stability is assessed by comparing the Hunt 1 and Hunt 2 score distributions across the same rubric dimensions. We compute the mean and variance of dimension scores across the three scored puzzles in each hunt, and report the per-dimension score difference between runs. If scores are stable (similar means and variance across dimensions), the panel's quality signal is robust to the specific design choices made at Levels 3–5 (answer word selection, mechanism type, micro-design): the panel is measuring a quality dimension that is invariant across the arbitrary choices that distinguish the two runs. If scores are variable — particularly if certain rubric dimensions show large between-run differences while others do not — the panel is sensitive to those dimensions' underlying design choices, indicating that those choices have genuine quality consequences despite being drawn from an apparently unconstrained design space.

The panel score analysis thus serves a dual purpose: it directly compares the quality of the two hunts as creative artifacts, and it provides an empirical test of the stability of the AI panel's quality signal across independent runs of the same pipeline — a question of practical importance for pipeline operators who rely on AI panel scores as a deployment gate.

4 Results

We report results at each of the five comparison levels described in Section 3.3, followed by the panel score stability analysis. Each subsection presents the raw comparison data, the relevant metric values, and a stability classification. Table 3 at the close of this section consolidates all classifications into a single reference.

4.1 Level 1: Structural Stability

Both hunts comply exactly with the SCOPE.md specification: five feeder puzzles plus one meta-puzzle, a casual solo audience, an estimated 2–3 hour solve time, and the Monk narrator voice (short sentences, present tense, no exclamation marks). These properties are stable across runs by construction: they were explicitly enumerated in the brief and the pipeline enforces compliance at the assignment stage.

Beyond the specified constraints, structural decisions diverge substantially. Hunt 1 employs a flat architecture: all five feeders reside in a single round titled “The Ages,” with no gating structure. Hunt 2 introduces a three-round gated architecture — *Dark Age* (P1–P2), *Castle Age* (P3–P4), *Imperial Age* (P5 + meta) — that mirrors the in-game age-advancement mechanic and distributes the difficulty arc across discrete thematic sections. The round count mismatch (1 vs. 3) represents a genuine architectural choice: Hunt 2's structure requires the solver to complete earlier rounds before accessing later puzzles, while Hunt 1's flat structure makes all feeders simultaneously available.

The meta mechanisms are categorically distinct. Hunt 1 employs a crossword-grid extraction: the five feeder answers (CASTLE, ON-AGER, LOOM, TOWER, PATROL) fill a grid, and six circled squares spell the meta answer WOLOLO. Hunt 2 employs an arithmetic index followed by anagram: for each feeder answer, the solver counts the vowels and adds one to obtain a positional index; the letter at that index is extracted; and the five extracted letters (R, A, G, I, L) are anagrammed to yield GRAIL. These belong to different mechanism

families (grid-navigation vs. rule-application-and-unscramble) and require different solver skills.

The Level 1 classification is therefore **variable** at the decision level (round count, meta mechanism type) and **stable** at the specification level (scale, narrator, audience, format, solve time). The stable properties are precisely those that were explicitly enumerated in the input brief; the variable properties are precisely those left to the generator's discretion.

4.2 Level 2: Pool Composition

Both runs produced a 15-candidate puzzle pool at Stage 3, with five designated slots containing three candidates each. The pools share identical cardinality but differ in their organizing principle. Hunt 1 organized the pool by AoE2 content domain: three candidates per slot, one slot each for civilizations, units, technologies, maps, and economy. Hunt 2 organized the pool by puzzle mechanism type, mixing domains within each slot and distributing candidates by generative family (chain-following, knowledge-identification, arithmetic, logic-grid, word-transformation).

The type distributions of the two pools diverge accordingly. Hunt 1's 15 candidates break down as: 5 knowledge-identification, 2 logic/deduction, 1 sequencing, 2 spatial/visual, 2 arithmetic, 2 proof-completion/gap-fill, and 1 upgrade-chain. Hunt 2's 15 candidates break down as: 4 knowledge-identification, 3 chain/path-following, 3 arithmetic/logic, 2 logic-grid, 1 word-transformation, 1 visual/spatial, and 1 sequencing. The two distributions share the knowledge-identification and visual/spatial types as their most prominent overlap, but the chain/path-following category (dominant in Hunt 2) is nearly absent from Hunt 1, and the arithmetic and proof-completion categories present in Hunt 1 have no strong parallel in Hunt 2's organization.

Computing Jaccard similarity over the five puzzle types represented in each hunt's final selection yields the most direct cross-run comparison. Hunt 1 selected: identification, logic/deduction, gap-fill, spatial/visual, and arithmetic. Hunt 2 selected: chain-following, path-following, spatial/visual, word-transformation, and logic-grid. The intersection is {spatial/visual}, a single shared type; the union spans nine distinct types. Jaccard similarity on the selected-type sets is therefore $1/9 = 0.11$ — well below the 0.4 stability threshold specified in Section 3.4.

The KL divergence between the two type distributions is approximately 0.42 nats, indicating moderate distributional dissimilarity (computed as $D_{KL}(H1||H2)$ over the aligned 8-type joint vocabulary, with smoothing value $1/30$ applied to types absent from one distribution). This falls below the 0.5-nat threshold specified in Table ?? but, taken together with Jaccard 0.11, consistently supports the VARIABLE classification: the two pools share minimal type overlap and are organized on categorically different principles.

The Level 2 classification is **variable**. Despite drawing from the same AoE2 content library, the two runs produced pools with substantially different type distributions (Jaccard 0.11 on selected types, KL divergence 0.42 nats), reflecting genuinely different generative organizing principles.

4.3 Level 3: Answer Word Assignment

The two answer word sets share no elements in common. Hunt 1's feeder answers are {CASTLE, ONAGER, LOOM, TOWER, PATROL}; Hunt 2's are {FORGE, MARCH, SIEGE, FAITH, RELIC}. Jaccard similarity on the two 5-word feeder sets is $0/10 = 0.00$. Including the meta answers (WOLOLO vs. GRAIL), no word appears in both hunts, and the 6-word Jaccard remains 0.00.

Word-length distributions differ as well. Hunt 1's feeders have lengths 6, 6, 4, 5, 6 (mean 5.4, range 4–6); Hunt 2's feeders are uniformly 5 letters each (mean 5.0, range 5–5). This uniformity in Hunt 2 is not incidental: the vowel-count index extraction mechanism requires answer words to have sufficient length to accommodate a positional index of up to four, and the anagram resolution is most tractable when feeder lengths are consistent.

Despite zero lexical overlap, domain-level semantic proximity is partially present. Three of the five feeder slots yield word pairs that are adjacent in the AoE2 lexical field: (LOOM, FORGE) — both name early-game technology structures in the production domain; (ONAGER, SIEGE) — both belong to the siege-warfare category; and (PATROL, MARCH) — both are military-movement verbs describing unit behavior. The remaining two pairs (CASTLE/FAITH and TOWER/RELIC) show no semantic proximity. This partial domain convergence, occurring in 3 of 5 slots at Jaccard 0.00, suggests that the AoE2 content library exerts gravitational pull toward particular semantic territories even when the specific lexical choices are unconstrained.

As a rough null model: if feeder answers were drawn uniformly at random from the AoE2 vocabulary (approximately 200 named items across civs, units, techs, and maps), the expected number of domain-adjacent pairs across the two 5-word sets would be approximately 0.8 under independence. The observed count of 3 is higher, though with $N=2$ this comparison is indicative rather than statistically conclusive.

The Level 3 classification is **variable** on exact strings (Jaccard 0.00) and **partially stable** on domain-semantic field (3 of 5 slots domain-adjacent).

4.4 Level 4: Extraction Mechanism

Table 2 shows the extraction mechanism family assigned to each puzzle slot in both hunts. The two hunts share a mechanism family in at most one of five feeder slots. Slot 1 presents the closest parallel: Hunt 1's civilization-bonus identification puzzle uses an index- N -from-answer- N rule (circle letter N from the N th answer), and Hunt 2's Blacksmith-chain puzzle uses the same abstract extraction principle (take letter N from blank N 's answer). The content substrate differs — civilizations and bonuses vs. upgrade-chain technology names — but the extraction logic is structurally identical. Slots 2 through 5, and the meta, are in categorically distinct mechanism families across the two runs.

Mechanism type Jaccard over the five feeder slots is $1/9 = 0.11$ (one shared family, nine distinct families in the union). The meta mechanisms are categorically distinct: grid-navigation and arithmetic-plus-anagram require different solver skills and produce structurally different artifacts. One notable convergence that is not captured in the slot-by-slot analysis: both hunts placed a first-letter or index-letter extraction mechanism in their civilization-bonus

Table 2: Extraction mechanism families by puzzle slot, both hunts. A “match” in the Overlap column indicates structural equivalence of the extraction rule, not content equivalence.

Slot	Hunt 1 Mechanism	Hunt 2 Mechanism
1	Knowledge-identification; indexed letter extraction	Chain-following; indexed letter extraction
2	Logic/deduction (tournament bracket)	Multi-domain tech identification
3	Gap-fill / tech-tree recall	First-letter from civ-bonus
4	Spatial/visual (dot-plot constellation)	Word transformation / word search
5	Arithmetic computation (A1Z26)	Logic grid (constraint satisfaction)
Meta	Grid-navigation (crossword)	Arithmetic index + anagram

identification puzzle (Hunt 1 slot 1 / Hunt 2 slot 3), suggesting that the combination of civ-bonus content and letter-index extraction constitutes a stable attractor in the AoE2 puzzle mechanism space. This convergence arises despite the two puzzles appearing in different slots, paired with different answer words, and deploying slightly different extraction sub-variants.

The Level 4 classification is **variable**. Five of six mechanism positions (including the meta) are in distinct mechanism families across the two hunts.

4.5 Level 5: Micro-Design

We compare the first-slot puzzles (Hunt 1 P1: *Civilization Bonus Matcher*; Hunt 2 P1: *The Blacksmith's Chain*) as the most structurally analogous pair available for micro-design analysis, given the Slot 1 mechanism-family convergence identified at Level 4.

Both puzzles use the extraction rule “take the N th letter from the N th answer” — the single stable sub-element at this level. All other micro-design dimensions diverge. Hunt 1's puzzle uses a list of eight bonus descriptions (six load-bearing, two chaff); Hunt 2's uses four ASCII chain diagrams with five total blanks (all five load-bearing). Hunt 1 includes a built-in cross-clue hint (“two of these bonuses concern infantry”); Hunt 2 provides no hints. The visual grammar is different: Hunt 1 presents each clue as a percentage-exact one-sentence bonus descriptor with age restriction and letter count; Hunt 2 presents each chain as a sequence of upgrade nodes connected by arrows, with each blank labeled by age, cost, and single-line effect. Flavor text density differs substantially: Hunt 1 opens with a single sentence; Hunt 2 opens with five sentences of Monk-voice narration that contextualizes the Blacksmith's setting before presenting the puzzle. The chaff mechanism in Hunt 1 (requiring the solver to identify which six of eight items matter) has no parallel in Hunt 2's clean all-load-bearing structure.

Despite these differences, both puzzles operate in the same difficulty band (both rated 2–3 out of 5), both test specific AoE2 knowledge with five to six load-bearing data points, and both comply with the Riven Standard: the extraction rule is stated in the flavor text and the solver can verify the answer letter-by-letter.

The Level 5 classification is **variable** overall, with one stable sub-element (extraction family: index- N -from-answer- N) and systematic divergence across content domain, visual grammar, structural complication (chaff vs. clean), and flavor text density.

4.6 Panel Score Stability

Hunt 1 and Hunt 2 were evaluated using the same C8 panel (Katz, Snyder, Young reviewer profiles) and the same 7-dimension, 45-point rubric. Hunt 1 scores are available for all five puzzles; Hunt 2 scores are available for the three authored puzzles (P1–P3).

Hunt 2's three scored puzzles produced a panel mean of 38.8/45 (86% of maximum), with individual scores of 39.0 (P1), 36.7 (P2), and 40.7 (P3). Normalized to percentage of maximum for cross-run comparability, Hunt 1's first three puzzle slots averaged 83% (24.3/30, 26.0/30, 24.0/30 on Hunt 1's equivalent 30-point rubric for those early runs). The between-run difference of 3 percentage points falls within the expected variability range across reviewer panels and is consistent with a **stable** pass/fail signal: every scored puzzle in both runs clears the passing threshold (71–73% of maximum).

Dimension-level analysis, normalized to a common 5-point scale, reveals that the aggregate stability conceals structured variation. Clarity (Hunt 1: 4.7/5; Hunt 2: 4.9/5) and Solvability (Hunt 1: 4.6/5; Hunt 2: 4.3/5) are stable across runs (differences of ± 0.3 or less). Elegance and Reading Reward both show a systematic upward shift of +1.2 points in Hunt 2 (Elegance: 3.1 vs. 4.3; Reading Reward: 3.3 vs. 4.5). A methodological note: the +1.2-point uplift in Elegance and Reading Reward should be interpreted with caution — Hunt 1 was evaluated on a 6-dimension equal-weight rubric while Hunt 2 used a 7-dimension weighted rubric with Elegance and Reading Reward double-counted, which partially confounds the comparison. This shift is not random variation: Elegance and Reading Reward were the two dimensions where Hunt 1 scored lowest, and these weaknesses were explicitly documented in Hunt 1's synthesis report. Hunt 2's puzzle designs — particularly the chain-diagram format of P1 and the civ-bonus acrostic format of P3 — produce higher structural elegance and stronger aha-moment quality by construction.

The panel score stability analysis thus yields a two-level conclusion: the *aggregate quality signal* is stable (both runs pass the quality gate, mean normalized scores within 4 percentage points), while *dimension-level scores* are variable in the specific dimensions most sensitive to micro-design choices. This supports using AI panel aggregate scores as a reliable quality gate independent of specific puzzle content choices, while cautioning that individual dimension scores reflect genuine design quality variation that the generator does not control deterministically.

4.7 Summary

Table 3 consolidates all classification results. The pattern is clear: specification-driven decisions are stable across runs; generative decisions are variable. The sole partial exception is domain-semantic gravity at Level 3, where zero lexical overlap coexists with partial domain-adjacency in 3 of 5 feeder slots — a finding that will be discussed further in Section ??.

5 Discussion

5.1 Specification Drives Convergence

The central finding of this study is not that domain knowledge constrains divergence — it is that *specification* does. Every design decision that is explicitly defined in the SCOPE.md brief converges

Table 3: Level-by-level stability summary. Each row represents one measurable design decision or metric, with the Hunt 1 and Hunt 2 values, the relevant metric, and the stability classification.

Level / Decision	Hunt 1	Hunt 2	Metric
<i>Level 1 — Structure</i>			
Scale (puzzles)	5 feeders + 1 meta	5 feeders + 1 meta	Exact count
Narrator voice	The Monk (identical rules)	The Monk (identical rules)	Categorical
Round count	1	3	Count: 1 v
Round gating	None (flat)	Gated by age	Architecture
Meta mechanism type	Crossword (grid)	Vowel-count index + anagram	Mechanism
<i>Level 2 — Pool</i>			
Pool size	15 candidates	15 candidates	Exact count
Selected type overlap	1 of 5 types shared	1 of 5 types shared	Jaccard =
Pool org. principle	By AoE2 domain	By mechanism type	Categorical
<i>Level 3 — Assignment</i>			
Feeder word overlap	CASTLE/ONAGER/LOOM/TOWER/PATROL	FORGE/MARCH/SIEGE/FAITH/RELIC	Jaccard =
Meta answer	WOLOLO	GRAIL	Exact: 0 m
Domain adjacency	—	—	3/5 slots p
Answer word lengths	4–6 (mean 5.4)	5–5 (mean 5.0)	Length dis
<i>Level 4 — Mechanism</i>			
Extraction families	5 distinct types	5 distinct types	Jaccard =
Slot-1 extraction rule	Index N from answer N	Index N from answer N	Family ma
Meta mechanism	Crossword	Arithmetic + anagram	Categorical
<i>Level 5 — Micro-design</i>			
P1 extraction sub-rule	Circle letter N from ans. N	Take letter N from blank N	Same fam
P1 content domain	Civilizations	Technologies	Different
P1 visual grammar	List format	Chain diagram	Different
P1 chaff structure	2 chaff items	0 chaff items	Structural
P1 flavor text	Sparse (1 sentence)	Dense (5 sentences)	Length/de
<i>Panel Scores</i>			
Pass/fail verdict	4/5 pass, 1 redesign	3/3 pass	Direction
Normalized avg. (slots 1–3)	83%	86%	$\Delta = +3\%$
Clarity dimension	4.7/5	4.9/5	$\Delta = +0.2$
Solvability dimension	4.6/5	4.3/5	$\Delta = -0.3$
Elegance dimension	3.1/5	4.3/5	$\Delta = +1.2$
Reading Reward dim.	3.3/5	4.5/5	$\Delta = +1.2$

perfectly across both runs: the scale (five feeders and one meta), the narrator voice (The Monk, with fixed stylistic rules), the domain (Age of Empires II), the audience (casual solo player), the format (printable Markdown), and the estimated solve time (2–3 hours) are identical across Hunt 1 and Hunt 2. Every design decision left to the pipeline's generative judgment — round structure, meta mechanism family, answer word selection, clue formulations — diverges substantially.

This is a stronger and more generalizable claim than the original hypothesis. The working hypothesis entering this study, drawn from design space theory [? ?], was that *domain knowledge* would be the primary stabilizing force: because both hunts must draw on

the same finite AoE2 knowledge base, the domain would constrain the output space and produce convergence. The results partially support this — domain semantic gravity (discussed below) produces thematic coherence at the word-set level even where exact words diverge — but the primary driver of convergence is not the domain. It is the specification. Decisions in SCOPE.md are stable; decisions not in SCOPE.md are variable, even when domain knowledge constrains the universe of valid answers to a relatively small set.

The practical implication for AI creative pipeline deployment is direct: to make a pipeline's outputs more predictable, specify more in the creative brief — not by narrowing the creative space or constraining aesthetic choices, but by reducing the number of decisions the pipeline must resolve autonomously. The practitioners' question "how much does the pipeline vary across runs?" should be reframed as "how much does my brief under-specify?" In both observed runs, every under-specified decision was variable; every explicitly specified decision was stable in both runs. This reframing converts the same-input divergence problem from a property of the AI system (its stochasticity) into a property of the human-AI interface (the specification density of the creative brief). These recommendations should be treated as provisional guidance supported by two runs in a single domain, to be confirmed with $N \geq 5$ replication before operational deployment.

Three candidate mechanisms could explain why specification produces convergence. First, a *hard-filter* mechanism: specified decisions eliminate options that would violate the brief, leaving fewer valid choices. Second, a *distributional-prior* mechanism: explicit specification shifts the model's token distribution toward brief-consistent options at each generation step. Third, an *attention-anchor* mechanism: specified elements act as attention weights that steer downstream generation toward coherent continuations. These mechanisms predict different failure modes and should be distinguished in future work with targeted ablations.

The five comparison levels are not causally independent: Level 3 (answer word assignment) causally constrains Levels 4 and 5 by determining which content is available for mechanism selection and clue formulation. A hunt that assigned GRAIL as its meta answer was necessarily constrained to produce a 5-letter anagram-compatible feeder set, which influenced mechanism choices at Level 4. This causal structure means that stabilising upstream decisions (Levels 1–3) has multiplicative stabilising effects downstream.

5.2 Domain Semantic Gravity

The Level 3 results reveal a phenomenon not anticipated by the original study design: *domain semantic gravity*. The exact answer word sets diverge completely (Jaccard = 0.00 on string match; no word appears in both hunts). Yet 3 of the 5 feeder slots exhibit domain-adjacent word pairs across the two runs: (LOOM, FORGE) in the production-technology slot, (ONAGER, SIEGE) in the siege-warfare slot, (PATROL, MARCH) in the military-movement slot. The AoE2 domain does not force convergence to specific words — it cannot, because the set of valid AoE2 vocabulary items is large and equally distributed across mechanism slots — but it creates a semantic basin of attraction. Both runs fall into the same region of the AoE2 lexical space even when they choose different specific words within that region.

We term this domain semantic gravity: an observed pattern in this case study, consistent with an apparent tendency ($N=2$) of a domain-grounded creative generator to select outputs that cluster in a semantic subspace defined by the domain's vocabulary and conceptual structure, even across independent runs that do not share a single exact token; replication is required to establish it as a general property. Domain semantic gravity is a *partial stabilizing force*: it does not produce stability at the string level, but it produces stability at the thematic level. This distinction is practically important for meta puzzle design. An experienced human constructor who needed exactly the word PATROL for a meta mechanism would not be helped by the knowledge that a different run would produce MARCH — both fall in the military-verb region of AoE2 vocabulary, but they are different strings and cannot be substituted for one another in a mechanism that depends on specific letter positions or word length.

This observation also refines the practical recommendation from the previous subsection. Domain semantic gravity implies that multiple runs of the same under-specified pipeline will tend to produce answer word sets that are thematically coherent with the domain but lexically incompatible with one another. Practitioners who need specific answer word properties (e.g., a particular word length, a specific letter pattern, or a particular semantic category for meta purposes) cannot rely on domain semantic gravity to deliver them; they must either specify those properties explicitly in the brief, or run the pipeline multiple times and apply human selection from the resulting word sets.

5.3 The Panel as a Stable Quality Signal

Panel score stability is among this study's most practically important findings. If AI panel scores were highly sensitive to the arbitrary design choices made at Levels 3–5 — answer words, extraction mechanisms, clue formulations — then the panel's quality signal would be unreliable as a deployment gate: it would pass some runs and fail others based on the incidental luck of word selection rather than genuine differences in puzzle quality. The results show the opposite. The pass/fail verdict is stable across all scored puzzles in both hunts (all puzzles in both runs that received full panel review achieved a passing score). The normalized aggregate score is stable across the two runs (83% and 86% of maximum, a 3-point gap within expected inter-run noise). Clarity and Solvability — the two dimensions most directly tied to the puzzle's structural correctness — are nearly identical across runs (+0.2 and -0.3 on a 5-point scale, respectively).

In both runs, the panel's quality signal appears robust to the specific generative choices made at Levels 3–5, suggesting a quality dimension that may be invariant to those choices ($N=2$; replication required). The panel penalizes structurally broken extraction, ambiguous clue formulations, and domain-knowledge failures; it rewards clean aha moments, fair deductions, and thematic coherence. These properties attach to the puzzle's structure and the solver's cognitive path, not to the specific answer word or the particular technology chain chosen to instantiate that structure. Both runs produced structurally sound puzzles, and both runs were rewarded similarly by the panel.

The systematic upward shift in Elegance (+1.2) and Reading Reward (+1.2) from Hunt 1 to Hunt 2 is the one panel-level finding that is not merely stable noise. This shift is consistent with, but not conclusively attributable to, a genuine mechanism-level improvement; the double-weighting of these dimensions in Hunt 2's rubric and the change in reviewer panel are confounds that cannot be disentangled from this case study alone. Hunt 2's P1 (The Blacksmith's Chain) adopts a chain-following structure with deterministic extraction that passes the Computer Test cleanly and achieves near-perfect Riven Standard compliance; Hunt 1's structurally comparable puzzle (Economy Puzzle, V) required a redesign flag. This illustrates a key distinction: in both observed runs, the panel was stable enough to confirm that both hunts are deployable-quality, but sensitive enough to detect genuine quality differences when they exist.

5.4 Limitations

Three primary limitations bound the conclusions of this study. First, the empirical base is $N = 2$ runs. Two independent runs provide a case study of same-input divergence; they do not provide a statistical characterization of its distribution. The divergence patterns observed here — Jaccard 0.00 at Level 3, 1/5 mechanism-type matches at Level 4, 3/5 domain-adjacent word pairs — could be atypical of the pipeline's general behavior. A robust characterization of divergence structure requires at minimum $N \geq 10$ runs across the same domain, from which distribution statistics (mean, variance, and tail behavior of each divergence metric) can be estimated.

Second, the clean-room isolation between Hunt 1 and Hunt 2 was enforced procedurally — the generating agent was instructed not to access Hunt 1 artifacts — rather than architecturally. In a multi-agent system with separate process boundaries, architectural isolation would guarantee that Hunt 1's generated outputs could not influence Hunt 2's generation even if instruction following were imperfect. The procedural approach is sufficient for a controlled study but introduces a residual confound that architectural isolation would eliminate.

Third, only three of Hunt 2's five feeder puzzles were fully authored and panel-reviewed at the time of analysis. The panel score comparison covers three puzzle slots per hunt, which is adequate to observe the stability pattern at the direction level but insufficient to characterize the full score distribution or to identify any outlier puzzles in Hunt 2 that might fall below the passing threshold. The full comparison awaits completion of Hunt 2's P4 and P5.

Future work should replicate this study with $N = 10$ runs across at least three domains (AoE2, music, science fiction) to assess whether the specification-drives-convergence finding generalizes across domain types with different knowledge density and vocabulary structure. Automated comparison scripts (currently a manual coding step at Level 4) would reduce the cost of large- N replication and enable near-real-time divergence monitoring in deployed creative pipelines.

6 Conclusion

This paper introduced the same-input divergence paradigm and applied it to characterize the structure of variation in an AI-driven puzzle hunt generation pipeline. Running the same AoE2 SCOPE.md brief, world knowledge base, and toolkit skill library twice under

clean-room conditions produced two complete hunts — Hunt 1 ("Wololo") and Hunt 2 ("Run 2") — whose outputs we compared at five levels of abstraction spanning round architecture, puzzle type selection, answer word assignment, extraction mechanism design, and individual clue formulations. The resulting empirical characterization contributes four findings to the literature on AI creative generation. First, same-input divergence in this pipeline is structured rather than random: stable and variable decisions partition cleanly into two groups, and that partition is principled rather than incidental. Second, and most importantly, the organizing principle of the partition is not domain knowledge but specification: decisions that are explicitly defined in the creative brief converge across runs; decisions that are left to the pipeline's generative judgment diverge, even when domain constraints narrow the valid option set. This overturns the original domain-constraint hypothesis and points toward specification density as the actionable lever for controlling pipeline predictability. Third, domain semantic gravity is a newly identified phenomenon in which a domain-grounded generator produces thematically coherent word sets across independent runs even when exact string overlap is zero, reflecting a semantic basin of attraction established by the domain's vocabulary and conceptual structure. Fourth, the AI panel's quality signal is moderately stable across the generative decisions that vary most (answer words, mechanisms, clue formulations), confirming that in both observed runs the panel measured structural and cognitive-path quality dimensions that were stable in both runs across the arbitrary choices at Levels 3–5, while remaining sensitive enough to detect genuine quality improvements at the mechanism level ($N=2$; replication required).

These findings translate directly into practical guidance for practitioners deploying AI creative generators in production pipelines. Decisions that must converge across runs — scale, format, narrator voice, audience, domain, tone — should be specified explicitly in the creative brief; in this pipeline and domain, explicitly specified decisions showed stable behaviour across both runs, suggesting a "run once" strategy may be appropriate for such decisions. For decisions where variety is desirable (answer word diversity, mechanism type mix), the under-specification should be deliberate, and practitioners should expect and welcome divergence across runs. Unspecified decisions diverged substantially in both runs, consistent with a "run and select" strategy for such decisions. For variable decisions that constrain downstream pipeline stages — most critically, answer words constraining meta mechanism design — the recommended practice is to run the pipeline multiple times, collect the resulting word sets, and apply human selection among options rather than trusting a single run. These practical recommendations are supported by two runs in a single domain and should be confirmed with $N \geq 5$ replication before operational deployment. The same-input divergence paradigm itself is a generalizable method that can be applied to any AI creative pipeline: run the pipeline twice from identical inputs, compare outputs at multiple levels of abstraction, identify which levels show stable behavior (specification-driven) and which show variable behavior (design-space-driven), and use this characterization to inform where in the pipeline human oversight is necessary and where a single AI run is sufficient.

Replication at $N = 10$ runs across three domains — AoE2, music, and science fiction — would establish whether the specification-drives-convergence finding generalizes to domains with markedly different knowledge density, vocabulary size, and structural constraints. Extending the comparison framework to include evaluative-language stability — asking whether the same evaluative frameworks appear in both runs' AI panel reviews, or whether the panel's conceptual vocabulary diverges across runs as the puzzle content

does — would connect the same-input divergence findings to the taxonomy of AI evaluative reasoning developed in the companion profile taxonomy paper [?], and would test whether the stability of panel *scores* is matched by a corresponding stability in panel *reasoning patterns*.

References